PROGRAMMING FUNDAMENTALS

Spring 2024

4th April

Ex-1

- Given the following function:

```
float func1(float x, float y, float z)
{
        return (x + y)/z;
}
```

- Write a call statement sending it values as per data types mentioned in an assignment statement. Get three values from the user. The assignment statement is included in main function.

Ex-2

- Your main function asks the user to enter number of hours worked (float) and the hourly rate (integer) for an employee
- Function calc_pay calculates the salary (float) and then deducts 10% as taxes from salary. It returns net_pay (float) to the main function.
- Main function displays the net_pay

Ex-3

- Ask the user to enter salary and number of children for an employee in main( ) function.
- The function, calc_edu_allowance is called when an employee has equal or greater than 2 children. The function call requires salary and it calculates and returns 15% of the employee's salary as education allowance.
- The main( ) function displays salary and education allowance.

Ex-4

- Write a program that allows input of two sides of a right triangle. It calls a function that calculates and returns the hypotenuse using Pythagorean theorem

(Hypotenuse = square root ((side1^2) + (side2^2))

Don’t forget to use:

#include <cmath>

To use sqrt() function

Sample Run..

D:\Books-ICT\C-Programs\Lec12-Q4.exe

```
Enter side1 of right angle triangle :   24.75

Enter side2 of right angle triangle =   38.2

Hypotenuse =   45.5171
--------------------------------
Process exited after 14.01 seconds with return value 0
Press any key to continue . . .
```

Ex-5

- An object in motion has kinetic energy calculated by :
- $KE = \frac{1}{2} mv^2$
- *m* is the object's mass in kilograms, and *v* is the object's velocity in meters per second
- Write a function named kineticEnergy that accepts an object's mass (in kilograms) and velocity (in meters per second) as arguments
- The function should return the amount of K.E that the object has
- Your main( ) function asks the user to enter values for mass and velocity

Let’s have Speed Programming competition today

Rules..

- Problem statement will be flashed on the slide.
- You are going to solve the problem by writing complete function prototype, main() function and function definition
- Whoever completes the task first posts the solution in the google classroom
- Once solution is posted, you will stop working on the current task and move on to the next
- Use the variable names and function names if given

Time for each task

- Minimum of 5 minutes
- Anyone who posts the solution before 5minutes or without getting permission (from me) will be disqualified

Let’s start

- Start a new source file
- Write the function body of the main() function
- Wait for the task to be shown on slide

Task-1

- Write the C++ code for a function that receives an integer from the user
- The function (divider_two) should divide the integer by 2 and then return the result to main() function in a float variable, my_ans

Task-2

- Your main() function asks the user to enter an integer.
- This integer is passed to a function, my_func1
- my_func1 calculates the square of the integer, and cube of the integer
- It returns the sum of the square and cube to the main() function
- Main() function displays the final answer (my_ans)

Task-3

- Your main() function uses a loop to enter scores (integers) of 5 tests (out of 10 each) of a student
- Then it calls the func_avg that calculates and returns the average
- If the student's average is greater than 8, the main() function displays "Well done" otherwise it displays "you must work harder"

Task-5

- Write the C++ code for a void function that receives an integer
- The function divides the integer by two and then displays the remainder
- Name the function my_remainder

Task-6

- Write the C++ code for a function that receives two float values
- The function divides the second value by first and returns the answer in a variable, my_q in main() function
- main() function displays the quotient
- Name the function my_quot

Task-7

- Write a program that determines which of a bank's four branches (Lahore, Karachi, Islamabad, Peshawar) had the greatest amount in accounts for a year
- You need two functions which are called by main()
- double get_Amt() is passed the name of a branch. It asks the user for last year's amount figure, validates the input, then returns it to be stored in an array in main()
- It should be called four times
- void find_max() is passed the four account amounts in an array and determines which is the maximum amount and prints the name of the branch along with its sales figure.
- *Input Validation: Do not accept amount less than 0*

Sample Run (Task-7)

D:\Books-ICT\C-Programs\Max_Bank_Bal.exe

```
Enter the Amount in Rupees in Lahore : 123
Enter the Amount in Rupees in Karachi : 345
Enter the Amount in Rupees in Islamabad : 908
Enter the Amount in Rupees in Peshawar : 544

Maximum Amount of 908  Rupees is in Islamabad
--------------------------------
Process exited after 21.01 seconds with return value 0
Press any key to continue . . .
```

Task-8

- Write a program that determines which of four major cities (Lahore, Karachi, Islamabad, Peshawar) had the least number of accidents in last month
- You need two functions which are called by main()
- int get_Accid() is passed the name of a city. It asks the user for last month's reported accidents, validates the input, then returns it to be stored in an array in main()
- It should be called four times
- void find_min() is passed the number of reported accidents in an array and determines which city has least accidents and prints the name of the city along with its number of accidents.
- *Input Validation: Do not accept accidents less than 0*

Sample Run (Task-8)

D:\Books-ICT\C-Programs\Accidents.exe

```
Enter the number of accidents in Lahore : 23
Enter the number of accidents in Karachi : 10
Enter the number of accidents in Islamabad : 100
Enter the number of accidents in Peshawar : 27

Minimum number of 10  accidents happened in Karachi
--------------------------------
Process exited after 13.14 seconds with return value 0
Press any key to continue . . .
```

Task-9

- Write a function named toss_coin
- The function should generate a random number in the range of 1 through 2
- If the random number returned is 1, the main() should display “It’s a Head”
- If the random number returned is 2, the main() function should display “It’s a Tail”
- Display the returned number in main() function for verification.
- Your main() also asks the user how many times the coin should be tossed and then calls the function accordingly using a loop

Sample Run (Task-9)

D:\Books-ICT\C-Programs\toss_coin.exe

```
How many times do you wish to toss the coin  :2
num = 1
It's a Head

num = 2
It's a Tail



--------------------------------
Process exited after 2.265 seconds with return value
Press any key to continue . . .
```